



A commentary by Elizabeth B. Gausden, MD, MPH, and Brian P. Chalmers, MD, is linked to the online version of this article.

SCIENTIFIC ARTICLES

RANDOMIZED CONTROLLED TRIALS

Oral Corticosteroids Reduce Pain After Total Knee Arthroplasty: A Higher Dose of Dexamethasone Effectively Controlled Pain During Motion

A Dose-Response Randomized Placebo-Controlled Trial

Supakit Kanitnate, MD, Yot Tanariyakul, MD, Pattaranatcha Charnwichai, PhD, Pimjai Anthanont, MD, Sudsayam Manuwong, MD, and Nattapol Tammachote, MD, MSc

Investigation performed at Thammasat University Hospital, Pathum Thani, Thailand

Background: Intravenous corticosteroids have been used to manage pain following total knee arthroplasty (TKA). Oral dexamethasone has good bioavailability and is suitable for ambulatory TKA. This study investigated the efficacy and dose-response relationship of oral dexamethasone in pain control after TKA.

Methods: In this randomized controlled trial, 120 Thai patients (mean, 68 ± 8 years; 86% female) undergoing primary TKA were allocated to 1 of 3 groups: 16 mg of oral dexamethasone (DEX-16), 8 mg of oral dexamethasone (DEX-8), or placebo. Dexamethasone or placebo was administered once daily, in the morning before surgery and the morning of the first 4 postoperative days. The primary outcome was pain at rest and during motion, measured over time. Linear mixed-effects modeling was used to compare outcomes between groups.

Results: Both DEX-8 and DEX-16 significantly reduced pain at rest within 48 hours postoperatively compared with placebo ($p < 0.05$). However, only DEX-16 significantly reduced pain during motion ($p < 0.05$). At 48 hours postoperatively, the DEX-16 group showed approximately 50% reductions in pain at rest ($p < 0.001$) and during motion ($p = 0.006$) relative to placebo.

Conclusions: Oral dexamethasone was effective in reducing pain after TKA, with a 16-mg dose providing superior pain relief during motion, compared with an 8-mg dose, within the first 48 hours. These findings support the use of oral dexamethasone as part of multimodal pain management, particularly in an ambulatory TKA setting.

Level of Evidence: Therapeutic Level I. See Instructions for Authors for a complete description of levels of evidence.

Multimodal analgesia is preferred to minimize postoperative pain and reduce the side effects of pain medications after total knee arthroplasty (TKA)^{1,2}.

Intravenous (IV) corticosteroids have been used to reduce postoperative pain, opioid consumption, and postoperative nausea and vomiting (PONV)³⁻¹⁴.

Disclosure: No external funding was received for this study. The Disclosure of Potential Conflicts of Interest forms are provided with the online version of the article (<http://links.lww.com/JBJS/I912>).

A **data-sharing statement** is provided with the online version of the article (<http://links.lww.com/JBJS/I914>).

J Bone Joint Surg Am. 2026 Jan 21 • 108(2): 120-7 • <http://dx.doi.org/10.2106/JBJS.25.00642>

Several studies have shown that corticosteroids are safe when used in appropriately selected patients^{5,15-17}. The use of corticosteroids for postoperative pain control in joint arthroplasty has gained popularity, with various routes of administration reported^{18,19,37}.

Dexamethasone is a corticosteroid with potent anti-inflammatory properties³. The bioavailability of dexamethasone administered orally is estimated to be approximately 80% relative to the IV route^{20,21}, and oral tablets may be better suited for ambulatory settings.

Few studies have examined the efficacy of oral corticosteroids for pain control after TKA. The optimal dosage, duration, and potential adverse effects of oral corticosteroids remain unclear. This study aimed to evaluate the efficacy of 2 different doses of oral dexamethasone, compared with placebo, in controlling pain after TKA.

Materials and Methods

Overview

We conducted a double-blinded, randomized controlled trial comparing the efficacy of 16 mg and 8 mg of oral dexamethasone with placebo for pain control following TKA. Participants were allocated in a 1:1:1 ratio. The study was approved by the institutional review board (MTU-EC-OT-6-097/62) and registered at ClinicalTrials.gov (NCT04244695). The trial was conducted between January 2020 and March 2023 at Thammasat University Hospital. All participants provided written informed consent.

Inclusion criteria were (1) primary unilateral TKA for osteoarthritis, (2) an age of 50 to 85 years, (3) American Society of Anesthesiologists (ASA) Physical Status I to III, and (4) ability to provide informed consent. Exclusion criteria included (1) uncontrolled diabetes (hemoglobin A1c of >7.0%), (2) a body mass index (BMI) of >40 kg/m², (3) inflammatory joint disease, (4) systemic corticosteroid use within 6 months previously, (5) a history of peptic ulcer hemorrhage, (6) cognitive or psychiatric impairment affecting pain assessment, (7) severe renal dysfunction (glomerular filtration rate [GFR] of <30 mL/min/1.73 m²) or hepatic dysfunction (Child-Pugh class C), (8) prior opioid abuse, and (9) allergy to any of the study medications.

Randomization

A total of 120 patients were randomized into 3 groups using computer-generated block randomization (blocks of 3). The DEX-16 group received 16 mg of oral dexamethasone daily for 5 days (a total of 80 mg), the DEX-8 group received 8 mg of oral dexamethasone daily for 5 days (a total of 40 mg), and the placebo group received placebo tablets. Allocation cards were

sealed in opaque envelopes and opened preoperatively by a study coordinator, who prepared medication packs. Each pack contained 4 tablets with identical appearance: (1) 4 tablets of 4 mg dexamethasone, (2) 2 tablets of 4 mg dexamethasone and 2 placebo tablets, or (3) 4 placebo tablets.

Intervention and Drug Protocol

Participants took 1 medication pack at 6:00 A.M. on the day of surgery and then each morning for 4 more days. Adherence was monitored by blinded evaluators (S.K., Y.T.). Both participants and evaluators were blinded to the group allocation.

All patients received pre-emptive analgesia consisting of 0.5 mg of lorazepam and 25 mg of pregabalin the night before and

the morning of surgery. Spinal anesthesia was performed using a 27-gauge needle with 10 to 15 mg of 0.5% bupivacaine. Prophylactic cefazolin (2 g) and tranexamic acid (750 mg) were administered intravenously 30 minutes before surgery. All procedures were performed under the supervision of the same senior surgeon (N.T.), with fellows and residents assisting in every case.

TKA was performed using a medial parapatellar approach with cemented cruciate-retaining implants and patellar resurfacing. A periarticular injection containing 100 mg of 0.5% bupivacaine, 30 mg of ketorolac, 5 mg of morphine sulfate, and 0.6 mg of epinephrine was administered after implantation. An additional 1.5 g of tranexamic acid was injected intra-articularly after wound closure.

Postoperative analgesia included 15 mg of IV ketorolac every 6 hours and 3 mg of IV morphine every 3 hours as needed during the first 48 hours. Patients were instructed to request IV morphine administration if visual analog scale (VAS) pain was >30 of 100 points. Patients also received oral naproxen (125 mg) and omeprazole (20 mg) twice daily, acetaminophen (1300 mg) every 8 hours, tramadol (50 mg) every 8 hours, and pregabalin (75 mg) and nortriptyline (25 mg) at bedtime. Patients were discharged on postoperative day (POD) 3 if they met the following criteria: ability to walk at least 10 m with a walker, pain that was well controlled with oral medications, no wound complications, and no urinary retention. Oral medications were continued for 2 weeks postoperatively. Apixaban (2.5 mg) was administered twice daily on PODs 1 through 10 for venous thromboembolism prophylaxis. Follow-up visits occurred at 2 and 6 weeks and 3 months postoperatively.

Assessments

Patients, surgeons, and evaluators were blinded to the treatment allocation. Postoperative symptoms were recorded by patients or caregivers using standardized forms, collected at discharge and follow-up.

“Oral dexamethasone was effective in reducing pain after TKA, with a 16-mg dose providing superior pain relief during motion, compared with an 8-mg dose, within the first 48 hours.”

The primary outcome was pain intensity at rest and during motion (knee flexion of $>45^\circ$), assessed using a 100-mm VAS with 0 indicating no pain and 100 indicating extreme pain. Patients were instructed to record their pain levels on pain record forms every 6 hours during hospitalization. They received 2 additional forms after discharge, with instructions to record their pain twice daily (at 8:00 A.M. and 8:00 P.M.) for 2 weeks, followed by weekly recording until 6 weeks. The forms were collected by a research assistant during outpatient visits at 2 and 6 weeks postoperatively.

Secondary outcomes included the rate of PONV, opioid consumption, fasting blood sugar (FBS), serum C-reactive protein (CRP) level, functional outcomes, and wound complications. PONV was classified as none, mild (antiemetics not required), or severe (antiemetics required). FBS was measured preoperatively and on PODs 1 and 2. CRP was measured on PODs 1, 2, and 14.

Functional outcomes were passive knee range of motion, the modified Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), and the Forgotten Joint Score (FJS). Range of motion was assessed at 24 and 48 hours, 2 and 6 weeks, and 3 months postoperatively. The modified WOMAC and FJS were recorded at 2 and 6 weeks and 3 months. Wound complications were assessed through the 3-month follow-up.

Statistical Analysis

The required sample size was calculated on the basis of a minimal clinically important difference (MCID) of 20% and a standard deviation estimated from a pilot study of 30 patients. Using an alpha level of 0.05 and power of 0.9, 32 patients per group were required. To account for a 20% dropout rate, 40 participants per group were recruited.

Statistical analyses were conducted using STATA 16.0 (StataCorp). Linear mixed-effects modeling was used to analyze VAS pain scores, opioid consumption, laboratory markers, and functional outcomes. When a global F-test indicated a significant difference among the groups, post hoc comparisons were performed using least significant difference (LSD) tests. The Fisher exact test was used for categorical variables. A p value of <0.05 was considered significant.

Results

A total of 120 Thai patients (mean age [and standard deviation], 68 ± 8 years; 86% female) were enrolled and randomized into 3 groups of 40 each. All patients received their assigned intervention on the morning of surgery (Fig. 1). Baseline demographics were similar across the groups ($p > 0.05$) (Table 1).

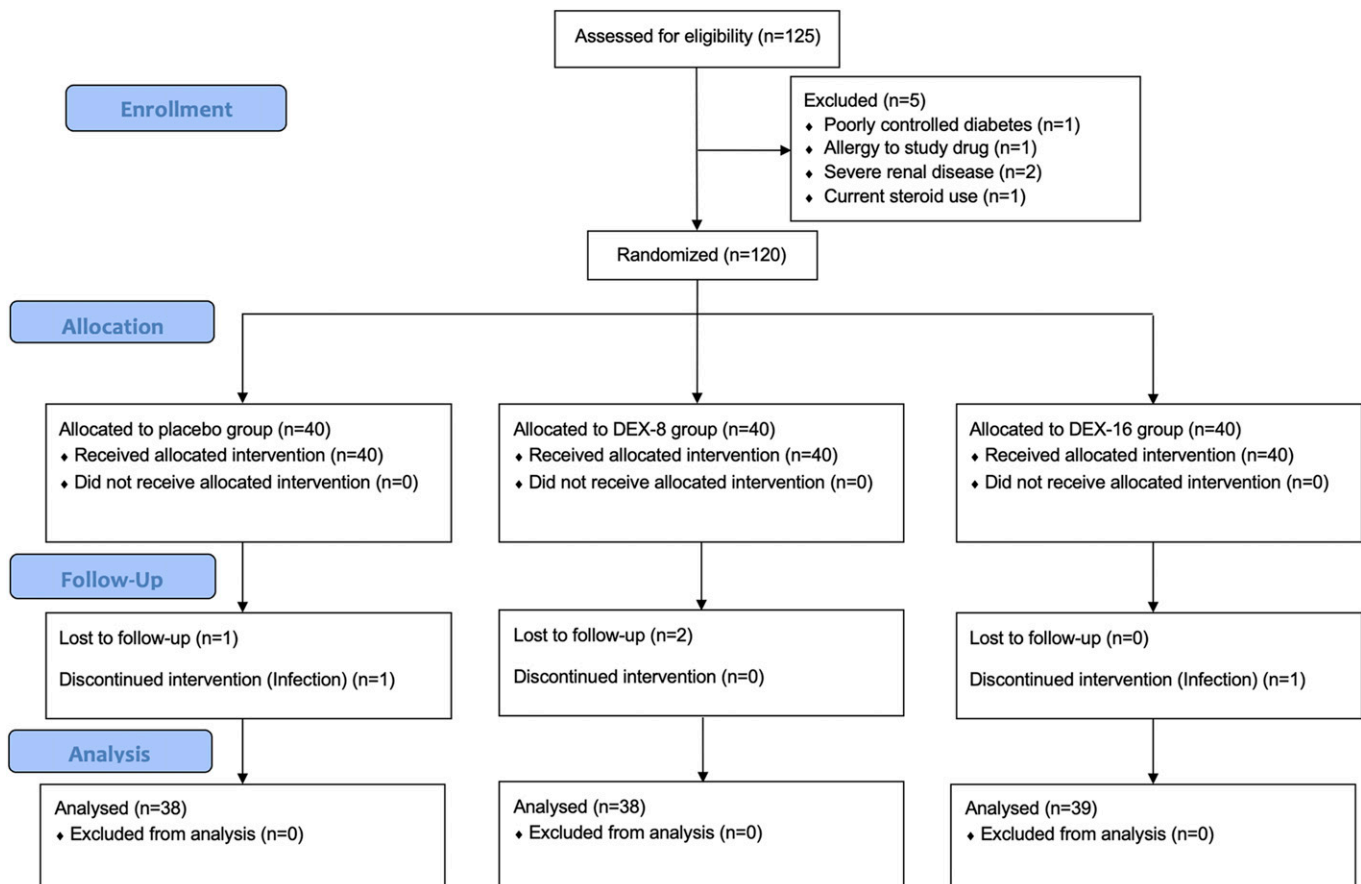


Fig. 1

CONSORT (Consolidated Standards of Reporting Trials) flow diagram of the study.

Table 1. Demographic Data*

Variable	DEX-16 Group (N = 39)†	DEX-8 Group (N = 38)†	Placebo Group (N = 38)†	P Value
Age† (yr)	66 ± 8	69 ± 8	68 ± 9	0.26
Sex, M/F	7/32	7/31	3/35	0.37
BMI† (kg/m ²)	27 ± 5	26 ± 4	27 ± 4	0.84
ASA classification (no. [%])				0.42
I	2 (5)	3 (8)	3 (8)	
II	35 (90)	28 (74)	31 (82)	
III	2 (5)	7 (18)	4 (11)	
Preop. range of motion† (deg)				
Flexion	120 ± 12	119 ± 15	118 ± 11	0.87
Extension	2 ± 5	1 ± 6	2 ± 6	0.79
Preop. modified WOMAC score† (points)	69 ± 16	71 ± 13	67 ± 14	0.54
Preop. limb alignment				
Varus/valgus	34/5	36/2	30/8	0.30
HKAA in varus knees† (deg)	172 ± 5	170 ± 8	170 ± 5	0.56
HKAA in valgus knees† (deg)	187 ± 6	183 ± 1	187 ± 9	0.78

*DEX-16 = 16 mg dexamethasone daily, DEX-8 = 8 mg dexamethasone daily, BMI = body mass index, ASA = American Society of Anesthesiologists, WOMAC = Western Ontario and McMaster Universities Osteoarthritis Index, HKAA = hip-knee-ankle angle. †The values are given as the mean ± standard deviation or as the number or number (percent) of patients.

The DEX-16 group showed significantly lower VAS pain scores during the first 48 hours postoperatively, both at rest and during motion, compared with the placebo group ($p < 0.05$) (Figs. 2-A and 2-B). Pain levels in the groups converged by POD 3. At 48 hours, the DEX-16 group had approximately 50% lower pain at rest ($p < 0.001$) and during motion ($p = 0.006$) compared with the placebo group. The DEX-8 group had significantly lower pain scores than the placebo group only at rest ($p < 0.05$). Detailed VAS pain scores are provided in the Appendix.

Rates of PONV were similar among the 3 groups ($p = 0.23$). Although the DEX-16 group demonstrated a trend toward reduced opioid consumption compared with placebo, the differences did not reach significance (Table 2).

The DEX-16 group had an approximately 12% higher mean FBS in the early postoperative period compared with the placebo group ($p < 0.05$). Conversely, its CRP levels were approximately threefold lower than those in the placebo group (Table 2).

Functional outcomes, including passive range of motion and the modified WOMAC and FJS, were similar among all groups at each follow-up ($p > 0.05$) (Table 3).

No superficial wound complications were observed. Two patients developed a periprosthetic joint infection (PJI): 1 in the DEX-16 group (treated successfully with debridement, antibiotics, and implant retention [DAIR] at 3 weeks after surgery) and 1 in the placebo group (requiring 2-stage revision after DAIR failure at 2 months postoperatively).

Discussion

This study demonstrated that oral dexamethasone effectively reduced pain after TKA. Both the 8 and 16-mg doses improved pain at rest during the first 48 hours compared with

placebo, while only the 16-mg dose significantly reduced pain during motion. Additionally, the oral dexamethasone lowered inflammatory markers and caused only a modest rise in blood glucose. Early functional outcomes were comparable across all groups within the first 3 months postoperatively.

Dexamethasone has been incorporated into multimodal analgesia protocols after TKA, and many studies have confirmed the analgesic efficacy of IV dexamethasone by comparison with placebo after TKA^{7-10,22}. A systematic review also found consistent benefits regardless of the administration route¹⁴. As ambulatory TKA becomes increasingly popular, oral options gain importance^{23,24}; however, oral corticosteroids have been investigated in only a few such studies^{18,19}. Cheng et al.¹⁹ administered 10 mg of oral prednisone daily for 14 days after TKA and reported pain relief through day 28. Shaw et al.¹⁸ compared 8 mg/day of oral dexamethasone (4 mg twice daily) with placebo over 4 days and found a reduction in pain scores that met the MCID only on POD 2. However, their patients also received 10 mg of IV dexamethasone preoperatively, which may have disguised the effect of the oral medication. In contrast, our study excluded IV dexamethasone and showed that multiple oral doses reduced pain within 6 to 48 hours postoperatively, a time frame that is in alignment with a previous report that peak pain occurred within the first 2 days following TKA and subsided thereafter²⁵.

To our knowledge, this is the first study comparing 2 oral dexamethasone doses in this context. Our findings indicate that 16 mg was more effective than 8 mg in reducing pain during motion in the early postoperative period. This dose-dependent effect is supported by IV studies. A meta-analysis reported that higher IV doses (>15 mg) reduced pain during motion more effectively than lower doses¹¹. Similarly, randomized controlled trials by Chan et al.¹² and Afshar et al.²⁶

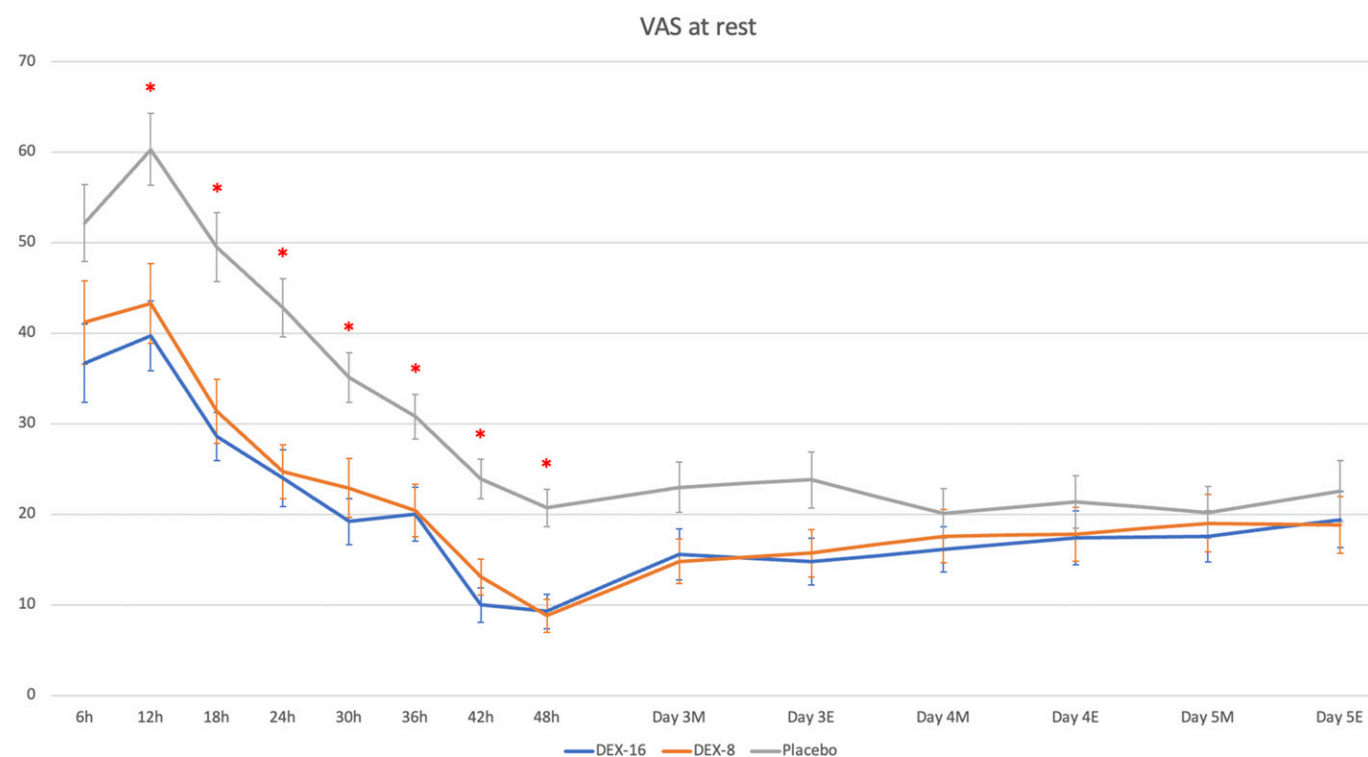


Fig. 2-A

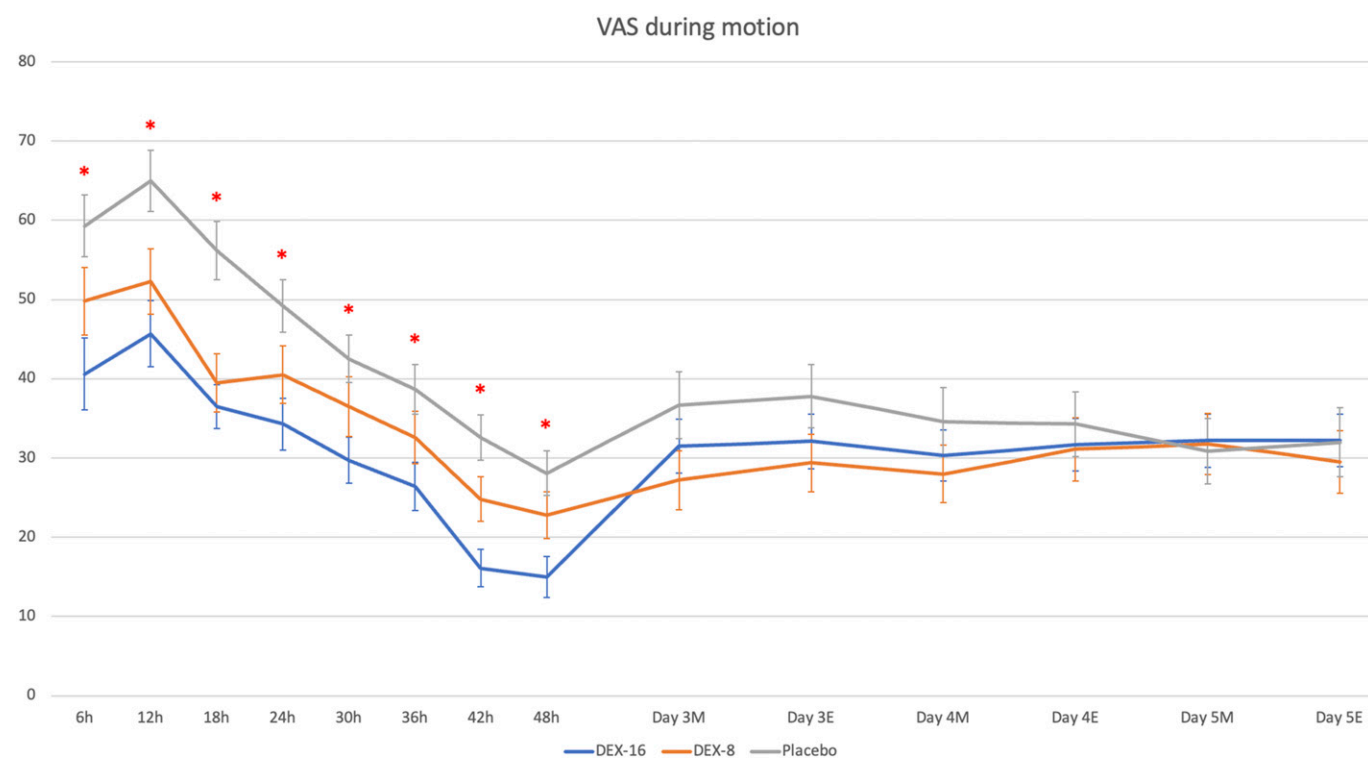


Fig. 2-B

Figs. 2-A and 2-B: The mean VAS score for pain at rest (**Fig. 2-A**) and during motion (**Fig. 2-B**) in the 3 treatment groups at each time point (PODs 0 through 5). The error bars indicate the standard error of the mean. *A significant difference among the 3 groups. M = morning, E = evening.

Table 2. Rate of PONV, Amount of Opioids Consumed, and Laboratory Results*

Variable	DEX-16 Group (N = 39)†	DEX-8 Group (N = 38)†	Placebo Group (N = 38)†	P Value			
				Overall	DEX-16 Vs. DEX-8	DEX-16 Vs. Placebo	DEX-8 Vs. Placebo
PONV				0.23			
None (<i>no. [%]</i>)	26 (67)	24 (63)	16 (42)				
Mild (<i>no. [%]</i>)	5 (13)	5 (13)	7 (18)				
Severe (<i>no. [%]</i>)	8 (21)	9 (24)	15 (40)				
Opioid consumption (<i>mg</i>)							
First 24 hours	1.9 ± 2.3	1.4 ± 2.3	3.2 ± 3.0	0.01	1.00	0.08	0.01
24-48 hours	0.9 ± 1.8	0.8 ± 2.1	1.5 ± 2.9	0.32			
Total	2.8 ± 3.5	2.2 ± 3.8	4.7 ± 4.9	0.02	1.00	0.13	0.02
FBS (<i>mg/dL</i>)							
Preop.	107 ± 13	107 ± 26	107 ± 19	0.99			
POD 1	136 ± 26	130 ± 24	122 ± 22	0.04	0.92	0.03	0.38
POD 2	133 ± 29	123 ± 24	117 ± 22	0.03	0.31	0.03	0.91
CRP (<i>mg/dL</i>)							
POD 1	23 ± 17	29 ± 25	47 ± 22	<0.001	0.81	<0.001	0.001
POD 2	68 ± 44	96 ± 55	189 ± 51	<0.001	0.049	<0.001	<0.001
POD 14	26 ± 26	24 ± 26	21 ± 24	0.81			

*DEX-16 = 16 mg dexamethasone daily, DEX-8 = 8 mg dexamethasone daily, PONV = postoperative nausea and vomiting, FBS = fasting blood sugar, POD = postoperative day, CRP = C-reactive protein. †The values are given as the mean ± standard deviation or as the number (percent) of patients.

Table 3. Functional Outcomes*

Variable	DEX-16 Group (N = 39)†	DEX-8 Group (N = 38)†	Placebo Group (N = 38)†	P Value
Range of motion				
Flexion (<i>deg</i>)				
24 hours	97 ± 16	89 ± 14	90 ± 19	0.19
48 hours	104 ± 14	101 ± 13	100 ± 16	0.53
2 weeks	121 ± 14	121 ± 12	119 ± 12	0.89
6 weeks	127 ± 11	129 ± 10	124 ± 23	0.43
3 months	127 ± 17	131 ± 9	132 ± 6	0.16
Extension (<i>deg</i>)				
24 hours	4 ± 5	5 ± 4	6 ± 4	0.37
48 hours	3 ± 4	3 ± 4	5 ± 4	0.18
2 weeks	1 ± 2	1 ± 2	2 ± 4	0.16
6 weeks	2 ± 4	2 ± 3	1 ± 2	0.50
3 months	1 ± 3	1 ± 2	1 ± 2	0.88
Modified WOMAC score (<i>points</i>)				
2 weeks	38 ± 13	39 ± 11	41 ± 14	0.70
6 weeks	20 ± 11	19 ± 11	22 ± 13	0.55
3 months	8 ± 8	12 ± 9	9 ± 5	0.15
Forgotten Joint Score (<i>points</i>)				
2 weeks	50 ± 37	43 ± 37	45 ± 37	0.69
6 weeks	67 ± 36	70 ± 31	70 ± 37	0.92
3 months	86 ± 25	85 ± 23	88 ± 18	0.79

*DEX-16 = 16 mg dexamethasone daily, DEX-8 = 8 mg dexamethasone daily, WOMAC = Western Ontario and McMaster Universities Osteoarthritis Index. †The values are given as the mean ± standard deviation.

both showed that 16 mg of IV dexamethasone yielded better outcomes than lower doses. Considering that oral dexamethasone has approximately 80% of the bioavailability of IV doses^{20,21}, 8 and 16-mg oral doses equal approximately 6.4 and 12.8-mg IV doses.

Opioid consumption was significantly lower only in the DEX-8 group compared with the placebo group, whereas the difference in the DEX-16 group did not reach significance. These results are inconsistent with previous studies that reported lower opioid use with higher IV dexamethasone doses²⁶. In our study, opioid use was generally low across all groups, likely due to the use of a multimodal analgesia protocol. Prior meta-analyses also found that perioperative dexamethasone reduced morphine use and the PONV rate after TKA^{6,11}.

Corticosteroids are known to exert strong anti-inflammatory effects. In our study, both DEX groups had lower CRP levels on PODs 1 and 2 than the placebo group, with a stronger effect observed in the higher-dose group. This supports prior findings that IV dexamethasone suppressed inflammatory markers after TKA^{3,10,27}. By 2 weeks, however, CRP levels normalized across groups.

Hyperglycemia is a well-recognized concern following systemic glucocorticoid use. Glucose levels of >180 mg/dL may impair leukocyte function and increase the risk of infection^{15,16,28,29}. However, large cohort studies and meta-analyses have not found an association of dexamethasone use with increased infection risk^{13,29}. Khair et al.³⁰ identified 137 mg/dL as the FBS threshold for elevated PJI risk. In our study, the DEX-16 group exhibited approximately 12% higher FBS, but levels stayed below the threshold. Consistent with earlier reports^{15,16,31}, transient hyperglycemia occurred without an increased PJI rate.

Hypothalamic-pituitary-adrenal (HPA) axis suppression by dexamethasone is another consideration. A study in healthy volunteers reported transient HPA suppression within 24 hours following a single 8-mg dose of IV dexamethasone³². However, another study found no significant suppression with perioperative doses of 2 to 10 mg³³. Prednisolone at doses of >5 mg/day for <2 weeks is generally low risk with respect to adrenal suppression^{34,35}. Even in patients with lymphoma receiving 100 mg/day of prednisolone for 5 days, adrenal suppression resolved within 30 days³⁶. Our regimen of 16 mg of oral dexamethasone daily (equivalent to approximately 80 mg of prednisolone) for 5 days also falls within this low-risk range.

This study has several limitations. First, the modest sample size (40 patients/group) may have limited our ability to detect rare complications such as PJI and may have been too small to detect differences in morphine consumption among the groups. Second, the short follow-up duration limited our

ability to assess long-term effects of the corticosteroid. Third, variability in the timing of drug administration relative to surgery could affect pharmacokinetics, although randomization helped mitigate this. Fourth, the multimodal analgesia approach—periarticular injection, acetaminophen, nonsteroidal anti-inflammatory drugs (NSAIDs), and opioids—may have confounded pain outcomes. However, we excluded adductor canal blocks and minimized the use of IV NSAIDs to better detect differences in pain levels between groups. Lastly, all patients were admitted for 2 days, as we collected outcomes including range of motion, FBS, and serum CRP level through 48 hours after surgery. While these findings may be extrapolated to patients in an ambulatory setting, further investigations in that specific group would be preferable.

In conclusion, oral dexamethasone effectively reduced early postoperative pain, specifically in the first 48 hours after TKA. A daily dose of 16 mg of oral dexamethasone offered greater benefit than 8 mg, particularly in the reduction of pain during motion. This approach is potentially more cost-effective and practical in an ambulatory setting than the IV route. Furthermore, it may also serve as an alternative for patients with contraindications to other analgesic drugs, such as NSAIDs.

Appendix

 Supporting material provided by the authors is posted with the online version of this article as a data supplement at <http://links.lww.com/JBJS/1913>. ■

NOTE: This study was internally supported by a Thammasat Postdoctoral Fellowship. The authors thank Cameron Hurst, PhD, for his help with the statistical analysis, and all participants for providing the data used in this study. We acknowledge the use of OpenAI's ChatGPT version 4.0 for assistance in improving the clarity and language of this manuscript; however, the content, analysis, and conclusions remain the sole responsibility of the authors.

Supakit Kanitnate, MD¹
Yot Tanariyakul, MD¹
Pattaranatcha Charnwichai, PhD¹
Pimjai Anthanont, MD²
Sudsayam Manuwong, MD³
Nattapol Tammachote, MD, MSc¹

¹Department of Orthopedics, Faculty of Medicine, Thammasat University, Pathum Thani, Thailand

²Department of Internal Medicine, Faculty of Medicine, Thammasat University, Pathum Thani, Thailand

³Department of Anesthesiology, Faculty of Medicine, Thammasat University, Pathum Thani, Thailand

Email for corresponding author: tammachotemd@gmail.com

References

1. Sah AP, Liang K, Sclafani JA. Optimal Multimodal Analgesia Treatment Recommendations for Total Joint Arthroplasty: A Critical Analysis Review. *JBJS Rev*. 2018 Jun;6(6):e7.
2. Summers S, Mohile N, McNamara C, Osman B, Gebhard R, Hernandez VH. Analgesia in Total Knee Arthroplasty: Current Pain Control Modalities and Outcomes. *J Bone Joint Surg Am*. 2020 Apr 15;102(8):719-27.
3. Tammachote N, Kanitnate S. Intravenous Dexamethasone Injection Reduces Pain From 12 to 21 Hours After Total Knee Arthroplasty: A Double-Blind, Randomized, Placebo-Controlled Trial. *J Arthroplasty*. 2020 Feb; 35(2):394-400.
4. Xu B, Ma J, Huang Q, Huang ZY, Zhang SY, Pei FX. Two doses of low-dose perioperative dexamethasone improve the clinical outcome after total knee

- arthroplasty: a randomized controlled study. *Knee Surg Sports Traumatol Arthrosc.* 2018 May;26(5):1549-56.
5. Hannon CP, Fillingham YA, Mason JB, Sterling RS, Hamilton WG, Della Valle CJ; AAHKS Anesthesia & Analgesia Clinical Practice Guideline Workgroup. Corticosteroids in Total Joint Arthroplasty: The Clinical Practice Guidelines of the American Association of Hip and Knee Surgeons, American Society of Regional Anesthesia and Pain Medicine, American Academy of Orthopaedic Surgeons, Hip Society, and Knee Society. *J Arthroplasty.* 2022 Sep;37(9):1684-7.
6. Hannon CP, Fillingham YA, Mason JB, Sterling RS, Casambre FD, Verity TJ, Woznica A, Nelson N, Hamilton WG, Della Valle CJ. The Efficacy and Safety of Corticosteroids in Total Joint Arthroplasty: A Direct Meta-Analysis. *J Arthroplasty.* 2022 Oct;37(10):1898-1905.e7.
7. Dissanayake R, Du HN, Robertson IK, Ogden K, Wiltshire K, Mulford JS. Does Dexamethasone Reduce Hospital Readiness for Discharge, Pain, Nausea, and Early Patient Satisfaction in Hip and Knee Arthroplasty? A Randomized, Controlled Trial. *J Arthroplasty.* 2018 Nov;33(11):3429-36.
8. Kim JK, Ro DH, Lee HJ, Park JY, Han HS, Lee MC. Efficacy of Systemic Steroid Use Given One Day After Total Knee Arthroplasty for Pain and Nausea: A Randomized Controlled Study. *J Arthroplasty.* 2020 Jan;35(1):69-75.
9. Salerno A, Hermann R. Efficacy and safety of steroid use for postoperative pain relief. Update and review of the medical literature. *J Bone Joint Surg Am.* 2006 Jun;88(6):1361-72.
10. Xu H, Zhang S, Xie J, Lei Y, Cao G, Pei F. Multiple Doses of Perioperative Dexamethasone Further Improve Clinical Outcomes After Total Knee Arthroplasty: A Prospective, Randomized, Controlled Study. *J Arthroplasty.* 2018 Nov;33(11):3448-54.
11. Lex JR, Edwards TC, Packer TW, Jones GG, Ravi B. Perioperative Systemic Dexamethasone Reduces Length of Stay in Total Joint Arthroplasty: A Systematic Review and Meta-Analysis of Randomized Controlled Trials. *J Arthroplasty.* 2021 Mar;36(3):1168-86.
12. Chan TCW, Cheung CW, Wong SSC, Chung AYF, Irwin MG, Chan PK, Fu H, Yan CH, Chiu KY. Preoperative dexamethasone for pain relief after total knee arthroplasty: A randomised controlled trial. *Eur J Anaesthesiol.* 2020 Dec;37(12):1157-67.
13. Li X, Xu G, Xie W, Ma S. The efficacy and safety of dexamethasone for pain management after total knee arthroplasty: A systematic review and meta-analysis. *Int J Surg.* 2018 May;53:65-71.
14. Fan Z, Ma J, Kuang M, Zhang L, Han B, Yang B, Wang Y, Ma X. The efficacy of dexamethasone reducing postoperative pain and emesis after total knee arthroplasty: A systematic review and meta-analysis. *Int J Surg.* 2018 Apr;52:149-55.
15. Jones IA, Wier J, Liu KC, Richardson MK, Yoshida B, Palmer R, Lieberman JR, Heckmann ND. Dexamethasone-Associated Hyperglycemia is Not Associated With Infectious Complications After Total Joint Arthroplasty in Diabetic Patients. *J Arthroplasty.* 2024 Aug;39(8S1):S43-52.e5.
16. Volkmar AJ, Schultz JD, Rickert MM, Polkowski GG, Engstrom SM, Martin JR. Dexamethasone Is Associated With a Statistically Significant Increase in Postoperative Blood Glucose Levels Following Primary Total Knee Arthroplasty. *Arthroplast Today.* 2023 Jan 3;19:101076.
17. Vuorinen MA, Palanne RA, Mäkinen TJ, Leskinen JT, Huhtala H, Huotari KA. Infection safety of dexamethasone in total hip and total knee arthroplasty: a study of eighteen thousand, eight hundred and seventy two operations. *Int Orthop.* 2019 Aug;43(8):1787-92.
18. Shaw JH, Wesemann LD, Banka TR, North WT, Charters MA, Davis JJ. The AAHKS Clinical Research Award: Oral Dexamethasone Following Total Knee Arthroplasty: A Double-Blind, Randomized Controlled Trial. *J Arthroplasty.* 2023 Jul;38(7)(Suppl 2):S15-20.
19. Cheng X, Wang Z, Zhang Y, Zhang X. Oral administration of prednisone effectively reduces subacute pain after total knee arthroplasty. *Orthop Traumatol Surg Res.* 2021 May;107(3):102770.
20. Duggan DE, Yeh KC, Matalia N, Ditzler CA, McMahon FG. Bioavailability of oral dexamethasone. *Clin Pharmacol Ther.* 1975 Aug;18(2):205-9.
21. Spoorbergen SM, Deneer VH, Grutters JC, Pulles AE, Voorn GP, Rijkers GT, Bos WJ, van de Garde EM. Pharmacokinetics of oral vs. intravenous dexamethasone in patients hospitalized with community-acquired pneumonia. *Br J Clin Pharmacol.* 2014 Jul;78(1):78-83.
22. Lei Y, Huang Q, Xu B, Zhang S, Cao G, Pei F. Multiple Low-Dose Dexamethasone Further Improves Clinical Outcomes Following Total Hip Arthroplasty. *J Arthroplasty.* 2018 May;33(5):1426-31.
23. Sershon RA, Ast MP, DeCook CA, Della Valle CJ, Hamilton WG. Advanced Concepts in Outpatient Joint Arthroplasty. *J Arthroplasty.* 2024 Sep;39(9S2):S60-4.
24. Nowak LL, Schemitsch EH. Trends in Outpatient Total Knee Arthroplasty From 2012 to 2020. *J Arthroplasty.* 2023 Jun;38(6S):S21-5.
25. Lebleu J, Pauwels A, Poilvache H, Anract P, Belbachir A. Severe Postoperative Pain in Total Knee Arthroplasty Patients: Risk Factors, Insights and Implications for Pain Management via a Digital Health Approach. *J Clin Med.* 2023 Dec 15;12(24):7695.
26. Afshar AM, Sharma SK, Hamidi SH, Marznaki ZH, Mudgal SK, Kamyari N, Khalilzad M. Comparison of Dexamethasone at Three Doses Administered Postoperatively for Improving Pain Control and Sleep Quality in Patients Who Underwent Total Knee Arthroplasty: A Triple Blind Randomized Controlled Trial. *J Arthroplasty.* 2025 Mar;40(3):658-64.
27. Nielsen NI, Kehlet H, Gromov K, Troelsen A, Husted H, Varnum C, Kjarsgaard-Andersen P, Rasmussen LE, Pleckaitiene L, Foss NB. High-dose dexamethasone in low pain responders undergoing total knee arthroplasty: a randomised double-blind trial. *Br J Anaesth.* 2023 Mar;130(3):322-30.
28. Richardson AB, Bala A, Wellman SS, Attarian DE, Bolognesi MP, Grant SA. Perioperative Dexamethasone Administration Does Not Increase the Incidence of Postoperative Infection in Total Hip and Knee Arthroplasty: A Retrospective Analysis. *J Arthroplasty.* 2016 Aug;31(8):1784-7.
29. Heckmann ND, Wang JC, Piple AS, Marshall GA, Mills ES, Liu KC, Lieberman JR, Christ AB. Is Intraoperative Dexamethasone Utilization Associated With Increased Rates of Periprosthetic Joint Infection Following Total Joint Arthroplasty? *J Arthroplasty.* 2023 Feb;38(2):224-231.e1.
30. Khair MM, Tan TL, Khair M, Maltenfort MG, Chen AF. Postoperative Blood Glucose Levels Predict Infection After Total Joint Arthroplasty. *J Bone Joint Surg Am.* 2018 Aug 15;100(16):1423-31.
31. Kebaish KJ, Wang JC, Piple AS, Wier J, Liu KC, Christ AB, Heckmann ND. Is Dexamethasone Administration During Total Hip and Knee Arthroplasty Safe in Diabetic Patients? *J Am Acad Orthop Surg.* 2025 Mar 1;33(5):242-52.
32. Elston MS, Conaglen HM, Hughes C, Tamatea JA, Meyer-Rochow GY, Conaglen JV. Duration of cortisol suppression following a single dose of dexamethasone in healthy volunteers: a randomised double-blind placebo-controlled trial. *Anaesth Intensive Care.* 2013 Sep;41(5):596-601.
33. El-Sibai K, Rajpal A, Al-Aridi R, Selman WR, Arafah BM. The impact of perioperative dexamethasone administration on the normal hypothalamic pituitary adrenal response to major surgical procedures. *Endocrine.* 2017 Oct;58(1):134-42.
34. Martin-Grace J, Tomkins M, O'Reilly MW, Sherlock M. Iatrogenic adrenal insufficiency in adults. *Nat Rev Endocrinol.* 2024 Apr;20(4):209-27.
35. Simpson H, Tomlinson J, Wass J, Dean J, Arlt W. Guidance for the prevention and emergency management of adult patients with adrenal insufficiency. *Clin Med (Lond).* 2020 Jul;20(4):371-8.
36. Owattanapanich W, Sirinvaravong S, Suphadirekku K, Wannachalee T. Transient adrenal insufficiency in diffuse large B cell lymphoma patients after chemotherapy with short-course, high-dose corticosteroids. *Ann Hematol.* 2018 Dec;97(12):2403-10.
37. Tammachote N, Seangleulur A, Kanitnate S. Lumbar Epidural Corticosteroid Injection Reduces Subacute Pain and Improves Knee Function in the First Six Weeks After Total Knee Arthroplasty: A Double-Blinded Randomized Trial. *J Bone Joint Surg Am.* 2018 Jun 6;100(11):950-7.